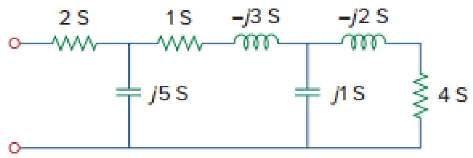
Chapter 9, Problem 69P

Problem

Find the equivalent admittance **Y**eq of the circuit in Fig. 9.76.

Figure 9.76:



Step-by-step solution

Step 1 of 8

Refer to Figure 9.76 in the textbook.

Draw the circuit diagram.

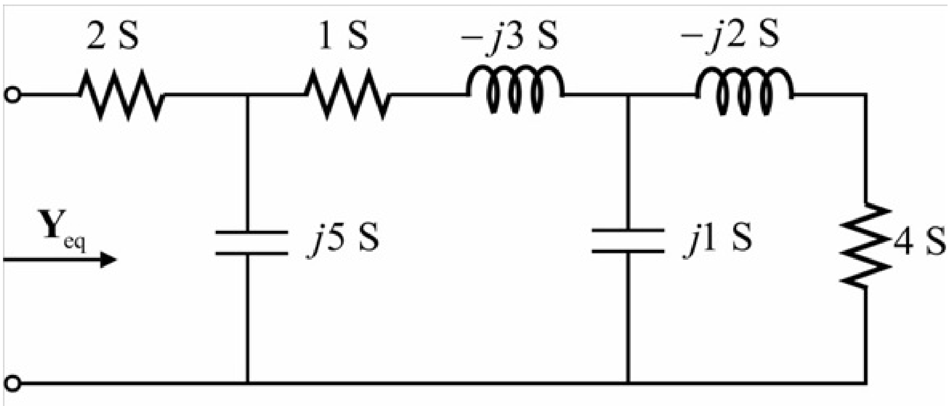


Figure 1

Step 2 of 8

From Figure 1, $-j2\ \text{S}$ and $4\ \text{S}$ are connected in series.

Calculate the equivalent admittance.

$$\frac{1}{\mathbf{Y}_1} = \frac{1}{-j2} + \frac{1}{4}$$
$$= \frac{(4)+(-j2)}{(4)(-j2)}$$
$$\mathbf{Y}_1 = \frac{-j8}{4-j2}$$
$$= 0.8 - j1.6\ \text{S}$$

Step 3 of 8

Draw the modified circuit diagram.

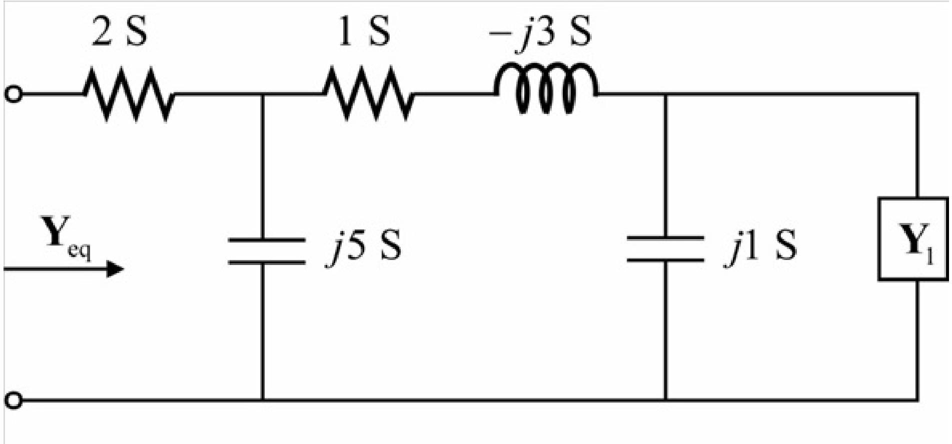


Figure 2

Step 4 of 8

From Figure 2, $j1$ S and $\mathbf{Y}_1$ are connected in parallel.

Calculate the equivalent admittance.

$$\begin{aligned}\mathbf{Y}_2 &= (j1) + \mathbf{Y}_1 \\ &= j1 + 0.8 - j1.6 \\ &= 0.8 - j0.6\end{aligned}$$

Draw the modified circuit diagram.

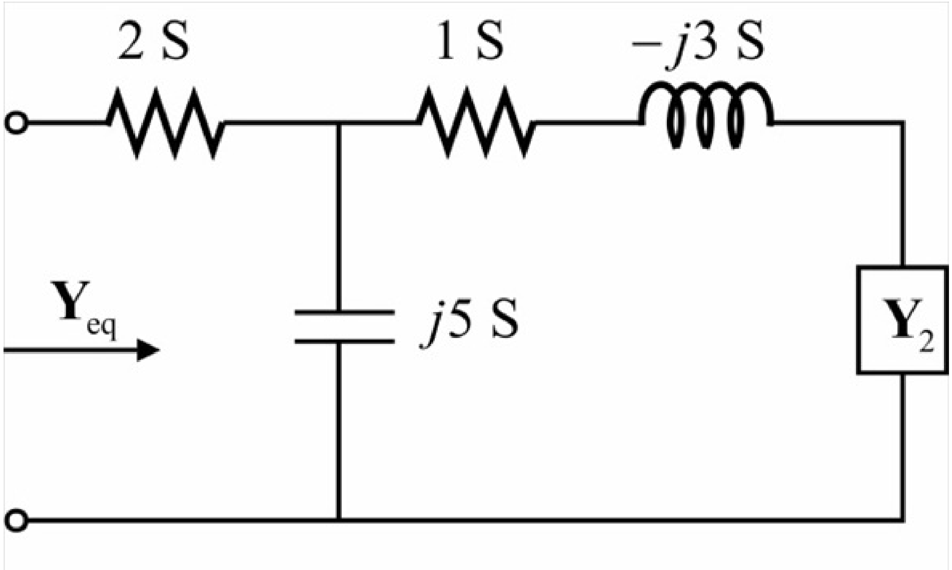


Figure 3

Step 5 of 8

From Figure 3, 1 S, $-j3\text{ S}$ and $\mathbf{Y}_2$ are connected in series.

Calculate the equivalent admittance.

$$\frac{1}{\mathbf{Y}_3}=\frac{1}{1}+\frac{1}{-j3}+\frac{1}{\mathbf{Y}_2}$$
$$=\frac{1}{1}+\frac{1}{-j3}+\frac{1}{0.8-j0.6}$$
$$=1+j0.334+0.8+j0.6$$
$$=1.8+j0.934$$

$$\mathbf{Y}_3=\frac{1}{1.8+j0.934}$$
$$=0.4378-j0.227\text{ S}$$

Step 6 of 8

Draw the modified circuit diagram.

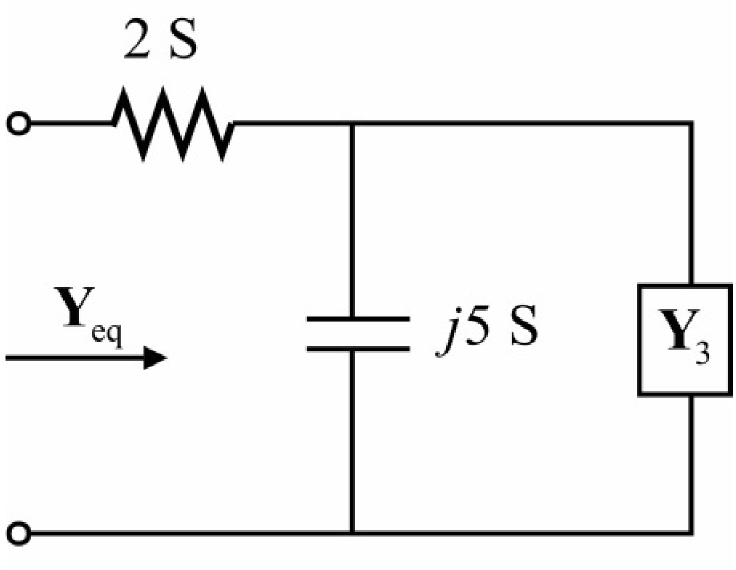


Figure 4

Step 7 of 8

From Figure 4, $j5\ \text{S}$ and $\mathbf{Y}_3$ are connected in parallel.

Calculate the equivalent admittance.

$$\begin{aligned}\mathbf{Y}_4 &= (j5)+\mathbf{Y}_3\\ &= j5+0.4378-j0.227\\ &= 0.4378+j4.773\ \text{S}\end{aligned}$$

Draw the modified circuit diagram.

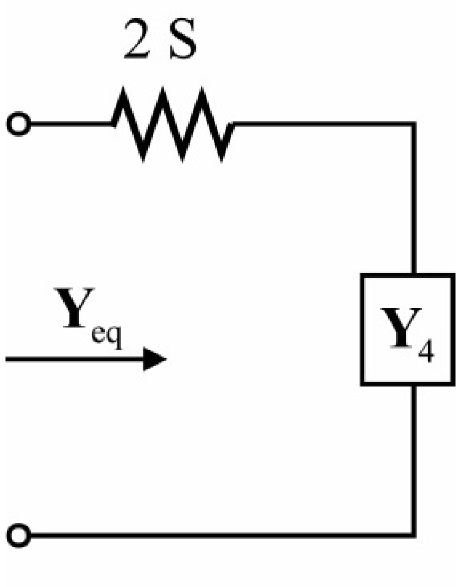


Figure 5

Step **8** of 8

From Figure 5, $2\ \text{S}$ and $\mathbf{Y}_4$ are connected in series.

Calculate the equivalent admittance.

$$\begin{aligned}\frac{1}{\mathbf{Y}_{eq}} &= \frac{1}{2}+\frac{1}{\mathbf{Y}_{eq}}\\ &= \frac{1}{2}+\frac{1}{0.4378+j4.773}\\ &= 0.5+0.019-j0.2\\ &= 0.519-j0.2\end{aligned}$$

$$\begin{aligned}\mathbf{Y}_{eq} &= \frac{1}{0.519-j0.2}\\ &= 1.66+j0.66\ \text{S}\end{aligned}$$

Thus, the equivalent admittance of the circuit is $\boxed{1.66+j0.66\ \text{S}}$.